

Programming in "C++"

Max Marks : 80

Min Marks : 27

Note : The Question Paper setter is advised to prepare unit-wise question with the provision of internal choice. Only Simple calculators allowed not scientific calculator.

UNIT - I: Language Fundamental

Overview of OOP: The Object Oriented paradigm, Basic concepts of OOP, Benefits of OOP, Object oriented languages, Application of OOP

Overview of C++: History of C++, **Data Types:** Built-in data types, User-defined data types, Derived data types. **Constants and Variables:** symbolic constants, Dynamic initialization of variable, Reference variable. Operators in C++. **Control Structures:** if-else, nested if-else, while, do-while, for, break, continue, switch, goto statement.

UNIT - II: Structure & Function

Structures: A Simple structure, defining a structure variable, Accessing structure's member, Enumeration data type.

Function: Function Declaration, Calling Function, Function Definition, **Passing Arguments to function:** Passing Constant, Passing Value, Reference Argument, Structure as argument, Default Argument.

Returning values from function: return statement, Returning structure variable, Return by reference. Overloaded Function, Inline Function.

UNIT - III: Object Classes and Inheritance

Object and Class, Defining the class and its member, Making an outside function inline, nesting of member function, array as class member, structure and classes.

Memory allocation: memory allocation for objects, new and delete operator, static data member, static member functions, object as function argument.

Constructor & Destructor: Null and default constructor. Parameterized constructor, Constructor with default argument, copy constructor, class destructors,

UNIT - IV: Pointers and Inheritance

Pointers: Introduction, & and * operator, pointer to object, this pointer, pointer to derived class.

Inheritance: Introduction to inheritance, Types of inheritance, function overriding, Constructor in Derived class. **Access specifiers:** public, private, protected.

UNIT - V: Polymorphism

Dynamic polymorphism: Virtual function, Pure Virtual Function, Abstract class.

Static Polymorphism: Operator keyword, overloading unary operators (++ (pre increment and post increment), --) using operator function, overloading binary operators (+, -, ==, >=, <=, +=, <, >, []), Friend function, Friend class, overloading binary operators using friend function.

RECOMMENDED BOOKS :

1. **Object Oriented Programming with C++ :** E. Balagurusamy, The McGraw-Hill
2. **Let Us C++:** Yesvant Kanetkar, BPB Publications
3. **The C++ Programming Language:** Bjarne Stroustrup, Addison Wesley.
4. **Object Oriented Programming in C++ :** Robert Lafore, Galgotia Publications.

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